ISAC-NET: Model-Driven Deep Learning for Integrated Passive Sensing and Communication

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***Abstract*— Wireless communication with the enormous demands of sensing ability have given rise to the integrated passive sensing and communication (IPSAC) technology. The main challenge of IPSAC is how to achieve high sensing and communication performance by integrating the passive sensing and communication demodulation. In this paper, we propose an integrated sensing and communication (ISAC) signal processing optimization scheme by jointly processing the pilot and data signals. To solve the optimization problem, we propose an ISAC signal processing algorithm based on iterative optimization, which alternates the passive sensing and channel reconstruction to realize target sensing. However, the hyper-parameter configuration of the iterative optimization algorithm influences the performance of target detection and communication demodulation. Recognizing this fact, we propose a model-driven ISAC network (ISAC-NET) that adopts the block-by-block signal processing method to improve the communication and sensing performance. The proposed ISAC-NET obtains suitable hyper-parameters by deep learning to guarantee the performance and convergence of communication and sensing signal processing. From the simulation results, ISAC-NET obtains better communication performance than the traditional signal demodulation algorithm, which is close to OAMP-Net2. Compared to the 2D-DFT algorithm, ISAC-NET demonstrates significantly enhanced sensing performance. In summary, ISAC-NET is a promising tool for the IPSAC systems.**

***Index Terms*— Deep learning (DL), integrated sensing and communication (ISAC), passive sensing.**

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I. Introduction

SENSING and communication systems have been researched separately for many years. Recently, there has been a growing interest in integrating both functions into a single system. This involves sharing the hardware and frequency resources by designing the integrated signals and signal processing algorithms, which are mentioned by integrated sensing and communication (ISAC) systems. There are many potential benefits to the integration of communication and sensing functions, such as improving the spectral efficiency, reducing the device size, cost [1], and power consumption, as well as enhancing the performance of both functions [2], [3], [4].

ISAC systems can be classified into two categories based on whether sensing is realized as active mode or passive mode [14]. For active sensing, specifically refers to mono-static sensing, the transmitter and receiver are co-located, which means that the receiver knows the transmitted signal in advance. For this reason, the active sensing receiver can recover the target parameters by using the transmitted signal and the received echoes. This is due to the fact that the transmitted signal interacts with the target and generates echoes that contain valuable information about the target, which can be processed by the receiver to extract the target parameters. However, active sensing requires ISAC transmitter and receiver to send and receive signals at the same time, which faces the high self-interference problem in the ISAC receiver. Compared to the active sensing, passive sensing has advantages in concealment, efficiency and security [15]. In passive sensing, specifically refers to bi-static sensing, the sensing receiver recovers the environment parameters using the received communication signals. The signal processing of passive sensing is similar to the channel estimation of wireless communications, which can be achieved through direct reuse of existing communication devices. Since passive sensing receives sensing echoes in another site far from the transmitter, it does not have the co-frequency co-time full duplex requirement in the receiver. Therefore, the integrated passive sensing and communication system is possible to be realized with minor adaptation to the existing communication systems, and has attracted much attention in the research fields of ISAC systems [15]. There are many use cases of passive sensing, such as WiFi-based indoor passive positioning. In all,

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TABLE I
ACTIVE SENSING VS. PASSIVE SENSING

	Active sensing	Passive sensing
Main differences	1) The receiver and transmitter ends are together 2) The receiver knows the transmitted signal in advance 3) Communication and sensing are carried out at different nodes	1) The receiver and transmitter ends are separated 2) The receiver doesn't know the transmitted signal in advance 3) Communication and sensing are carried out at the same node
Sensing algorithms	Estimate target information from echo signal directly [5]–[9]	Demodulate communication information first, and then extract target information from echo signal [10]–[13]
Typical scenarios	1) Active sensing of drone swarms 2) Uplink and downlink active sensing of base station environment	1) WiFi sensing of indoor environment 2) Uplink and downlink passive sensing of base station environment

there are two main differences between active sensing and passive sensing:

- The active sensing receiver has prior knowledge of the transmitted signal while the passive sensing receiver does not. Which means that the receiver of active sensing knows the transmitted signal in advance while the passive sensing receiver does not.
- For integrated active sensing and communication, sensing and communication demodulation are processed on different nodes, and the discrete ISAC signal processing algorithm can meet the requirements of this scenario [15]. However, for integrated passive sensing and communication, sensing and communication demodulation are processed at the same node, and the error of communication demodulation and sensing results will affect each other. In this case, the traditional discrete design of ISAC signal processing algorithm cannot meet the requirements of this scenario.

The comparison details of active and passive sensing are shown in TABLE I.

For integrated passive sensing and communication systems, sensing and communication are processed at the same node. Since the transmitter and receiver in passive sensing are separated, the receiver does not have prior knowledge of the transmitted signal, which makes it difficult to extract the target parameters from the received signal. Passive sensing based on pilot signal can obtain the target parameters by transmitting fixed pilot signals and processing the echoes [8], [16]. For pilot signal based passive sensing, the receiver knows the transmitter signal in advance, which is similar to active sensing. Therefore, pilot signal based passive sensing can be realized by traditional active sensing algorithms. There have been many signal processing algorithms for joint sensing and communications [17], which can be classified into five main types.

- **Periodogram such as 2D-DFT [5], [6]**: These methods are simple and easy to implement, but require a full set of continuous samples in each domain, which may not always be satisfied.
- **Maximal likelihood estimation [7]**: These methods are suitable for low-dimension signals, but require searching to find the solutions, which has the high complexity.
- **Subspace methods such as multiple signal classification (MUSIC) [18]**: These methods can achieve very high resolution and can deal with the off-grid estimation. Nonetheless, the complexity of MUSIC is high because of singular value decomposition.
- **On-grid or off-grid algorithm such as compressive sensing (CS) and atomic norm minimization [8]**: These methods are flexible and do not require consecutive samples. However, it cannot obtain high performance when the number of parameters to be estimated is large.
- **Tensor based algorithms [9]**: There methods can reduce computational complexity and provide capability in high-dimension formulation and estimation, but need to be combined with other algorithms such as CS.

However, since the pilot signal only occupies part of the time and frequency resources of the transmit signal, the above mentioned pilot signal based passive sensing cannot fully utilize the signal resources to obtain higher sensing performance.

In comparison with the pilot signal based passive sensing, some studies focus on recovering the communication information and target parameters utilizing data signals, which enables to provide abundant time-frequency resources. Distinguished from the pilot signal based passive sensing, the receiver cannot obtain the data signal in advance. Typically, the modulated communications symbols are first recovered, and are further utilized to obtain the target parameters. Since the recovery of modulated communication symbols are processed in the condition of noise and interference, the symbols cannot be totally recovered correctly. Therefore, how to achieve passive sensing with demodulation errors is a major challenge in the data signal based passive sensing. There are some algorithms are proposed to solve this problem.

- **LoS-based passive sensing [10], [11]**: Berger et al. proposed an algorithm by demodulating the transmit symbols using the line of sight (LoS) path signals with strong power, and extracted the non-LoS (NLoS) echoes reflected by the targets [10]. After demodulating the transmit symbols, traditional sensing algorithms [5] can be utilized to recover the target parameters. The performance of data signal based passive sensing in ISAC systems has been analyzed by Li et al. [11]. LoS path signal interference is one of the key factors limiting the performance of the above algorithms.
- **Passive sensing based on signal reconstruction [13]**: Pei et al. proposed a direct wave suppression method based on signal reconstruction that can effectively suppress the direct path interference and stimulates the processing of target detection.

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- **Passive sensing based on space-time interference suppression [12]**: The space-time interference suppression algorithm was proposed to enhance the performance of multipath clutter, strong interference, and matched filtering detection performance under Doppler strong clutter.

Essentially, the existing ISAC signal processing algorithms [12] treat communication and sensing as separate and independent processes. The target delay and Doppler information obtained by passive sensing are not used for communication demodulation, and the demodulation communication information is not used to improve the estimation accuracy of target parameters through passive sensing, resulting in weak improvement of the performance for communication and sensing. In summary, the main challenge of the integrated passive sensing and communication is: How to jointly process the pilot signal and data signal in order to simultaneously improve the performances of target detection and communications demodulation.

To address the above challenge, this paper proposes an ISAC signal processing optimization problem which jointly processes the pilot and data signals. On the one hand, both pilot and data signals are used for passive sensing, which contains abundant signal resources, thus improves the sensing accuracy. On the other hand, more accurate sensing results contribute to the accurate acquisition of channel information, thus improves the communication performance. Compared with the separated communication and sensing systems, the ISAC signal processing optimization problem requires simultaneous recovery of the channel state information, data information, and environment targets information. Therefore, the ISAC signal processing optimization problem is non-convex and difficult to be solved directly.

In order to solve the proposed ISAC optimization problem efficiently, we consider adopting the idea of iterative optimization [6] and propose an ISAC signal processing algorithm based on iterative optimization, which carries out the passive sensing and communication demodulation alternately. The iterative optimization algorithm proposed in this paper is divided into three modules. The first module is the improved denoising-based approximate message passing (D-AMP) algorithm, which demodulates the transmitted signal data from the received signal. The second module is the 2D-DFT passive sensing algorithm, which extracts the environmental target parameter by comparing the transmitted and received signal. The third module is the ISAC channel reconstruction algorithm, which reconstructs the ISAC channel based on the sensing results.

Although the iterative optimization algorithm enables to improve the sensing and communication performance to a certain extent, the hyper-parameter configuration of the iterative optimization algorithm influences the performance. Improper hyper-parameter configuration results in a degradation to the performance and convergence time. In recent years, model-driven deep learning (DL) signal processing algorithms are gradually gaining attention to obtain suitable hyper-parameters, which was discussed in [19]. This approach involves explicitly incorporating domain knowledge and prior information into the model, which improves the robustness and accuracy under error conditions [20]. A general model-driven DL architecture, called ComNet [21], combines DL with expert knowledge in wireless communications. ComNet uses neural network to expand the modules in traditional processing methods, and obtains more application-specific hyper-parameters through training, which reduces the computational complexity and improves the processing performance compared with traditional methods [19]. However, these works are only designed for communication systems without considering the requirement of sensing processing in ISAC systems.

Against the above backdrop, we further propose an ISAC network (ISAC-NET) to realize the data transmission and sensing at the same time. The model-driven ISAC-NET combines the conventional alternating optimization and deep unfolding into a neural network. The ISAC-NET considers the key steps of ISAC signal processing, and designs three modules, including the passive sensing module, signal detection module and channel reconstruction module. In ISAC-NET, communication data demodulation and target parameter estimation are interlaced and interactively optimized to achieve the optimal performance of both functions. Moreover, this paper has carried out simulation verification. The simulation results verify that the ISAC-NET achieves better communication and passive sensing performance than traditional communication and sensing algorithms.

The contributions of this paper are summarized as follows:

1) In the system level, we propose an integrated passive sensing and communication signal processing scheme for combining pilot and data signals. Compared with the independent communication and sensing signal processing algorithms, the proposed scheme of joint processing of pilot and data signals obtains higher performances in sensing accuracy and communication demodulation, by utilizing more time and frequency resources.
2) In the aspect of signal processing, we propose an ISAC signal processing algorithm based on iterative optimization, which contains the improved D-AMP algorithm, the 2D-DFT passive sensing algorithm and the ISAC channel reconstruction algorithm. Different from existing signal processing algorithms that first demodulate the transmitted symbols and then obtain passive sensing results by using the demodulated symbols, the proposed algorithm obtains passive sensing results and communication demodulated symbols simultaneously.
3) In the architecture of DL network, we put forward the model-driven ISAC-NET that adopts the block-by-block signal processing method to improve the performances of communication and sensing. Suitable hyper-parameters are obtained in the ISAC-NET, which guarantees the performance and convergence time of the signal processing.
4) In performance analysis and evaluation, we analyze and evaluate the performance improvements of the ISAC-NET in range, velocity estimation, and communication signal detection through simulation. The simulation results show that the ISAC-NET achieves better communication and passive sensing performance.

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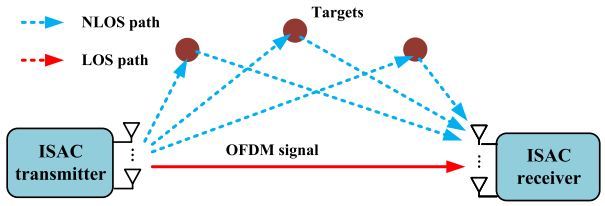


Fig. 1. Passive sensing and communication.

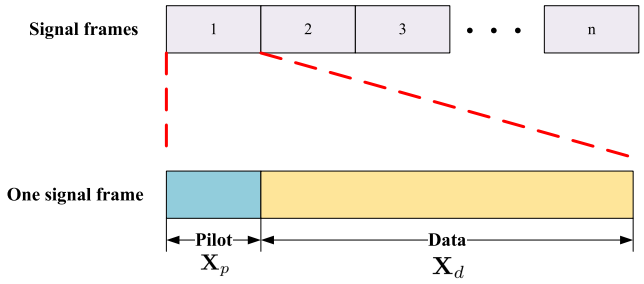


Fig. 2. Signal frame of passive sensing and communication.

In terms of communication, ISAC-NET obtains better communication performance than the traditional signal demodulation algorithm, and has close performance to that of OAMP-Net2 [22]. Compared to the 2D-DFT based sensing algorithm, ISAC-NET demonstrates significantly enhanced sensing performance and approaches the sensing performance without communication demodulation errors.

The rest of this paper is organized as follows: Section II describes the system model of integrated passive sensing and communication. The design of ISAC-NET is introduced in Section III and IV. The performance improvements in sensing and communication are analyzed and simulated in Section V. Section VI concludes the paper.

The following notations are used throughout the paper: Vectors and matrices are denoted by boldface small and capital letters. The transpose, complex conjugate, Hermite, inverse, and pseudo-inverse of the matrix $\mathbf{A}$ are denoted by $\mathbf{A}^{\mathrm{T}}$, $\mathbf{A}^{*}$, $\mathbf{A}^{\mathrm{H}}$, $\mathbf{A}^{-1}$ and $\mathbf{A}^{\dagger}$, respectively. $\mathrm{diag}(\mathbf{x})$ is the operation that generates a diagonal matrix with the diagonal elements to be the elements of $\mathbf{x}$. $\otimes$ is the Kronecker product operator. $\odot$ is the Hadamard product operator. $\mathbf{A}(m,n)$ denotes the element in row m, column n of the matrix $\mathbf{A}$. $\mathcal{S}(\mathbf{A}_1,\mathbf{A}_2)$ denotes the set of $\mathbf{A}_1$ and $\mathbf{A}_2$. $P(\mathbf{A})$ denotes the probability of $\mathbf{A}$.

II. System Model

A. Integrated Passive Sensing and Communication

We begin by introducing the multiple-input multiple-output (MIMO) system model of the proposed integrated passive sensing and communication system, as shown in Fig. 1. In this system, the ISAC receiver receives the multipath signals transmitted from the ISAC transmitter. Among these multipath signals, the LoS path signal is not reflected by the target and directly reaches the receiver, while the NLoS signals are reflected from the targets. The communications information is conveyed both by the LoS and NLoS signals, while the target parameters is only embedded in the NLoS signals. The task of the ISAC receiver is to recover the transmitted communication information and radar targets from the received multipath signals. In the sequel, we formulate the transmit signal model and the receive signal model in Section II – B and Section II – C, respectively.

B. Transmitted Signal Model

Assume that the OFDM symbols transmitted on different transmitting antennas are the same. Then, the transmit OFDM signal can be expressed as [5]

$$s(t)=\sum_{m=0}^{M-1}\sum_{n=0}^{N-1}x\left(m,n\right)\cdot e^{j2\pi(f_c+n\Delta f)t}\mathrm{rect}\left(\frac{t-mT}{T}\right), \tag{1}$$

where M is the number of OFDM symbols, N is the number of subcarriers, m is the index of OFDM symbol, n is the index of subcarrier, $x(m,n)$ is the complex modulation symbol, the rectangle function is $\mathrm{rect}\,(t)=1$ for $0\le t<1$, f_c is the carrier frequency, Δf denotes the subcarrier spacing. The duration of each OFDM symbol is $T=T_p+T_c$, where T_p is the elementary symbol duration and T_c is the guard interval. If T_c is larger than the maximum delay of the multipaths, the inter-symbol interference can be eliminated.

C. Received Signal Model

Assume that the ISAC transmitter and receiver have N_t and N_r antenna elements, respectively. As shown in Fig. 2, the ISAC signal $\mathbf{X}$ consists of the pilot signal $\mathbf{X}_p$ and the data signal $\mathbf{X}_d$. The pilot signals are used for channel estimation and initial passive sensing, while the data signals are utilized to recover the conveyed information. Typically, the pilot signal is a reference signal known to the receiver, such as the demodulation reference signal or position reference signal. Since both the pilot signal and the data signal occupy a part of time duration, there is a trade-off between the occupation ratio between the pilot and the data signal. On the one hand, increasing the length of pilot leads to a higher estimation accuracy of CSI, and thus achieves better sensing performance. On the other hand, using long pilot signal reduces the communication capacity because the pilot signal does not carry communication information. The true communication capacity can be expressed as [23]

$$\begin{aligned}C_t&=\frac{\tau_d}{\tau_d+\tau_p}B\log(1+SNR)\\&=(1-w_p)B\log(1+SNR)\end{aligned} \tag{2}$$

where B is the system bandwidth, τ_d is the length of the data signal, τ_p is the length of the pilot signal, $w_p=\frac{\tau_p}{\tau_p+\tau_d}$ is the length ratio of the pilot signal. Therefore, there are a trade-off between the sensing accuracy and the true communication capacity because of the length ratio of the pilot signal. In the design of commercial mobile communication systems, the length of the pilot signal has been widely studied, and different pilot parameters are set in the standards for different scenarios.

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In 5G NR standard [23], 1 or 2 OFDM symbols are typically set as the pilot signal among 14 OFDM symbols in each time slot. Without loss of generality, this paper adopts the typical pilot length ratio setting in 5G NR, i.e., 1/14, in the analysis and numerical simulation parts.

Then, the transmitted symbols can be rewritten in a matrix form given by

$$\mathbf{X}' = \begin{bmatrix} x(0,0) & \cdots & x(0,N-1) \\ x(1,0) & \cdots & x(1,N-1) \\ \vdots & \ddots & \vdots \\ x(M-1,0) & \cdots & x(M-1,N-1) \end{bmatrix}, \tag{3}$$

and the received signal matrix from the k-th receiving antenna can be expressed as

$$\mathbf{Y}'_k = \begin{bmatrix} y_k(0,0) & \cdots & y_k(0,N-1) \\ y_k(1,0) & \cdots & y_k(1,N-1) \\ \vdots & \ddots & \vdots \\ y_k(M-1,0) & \cdots & y_k(M-1,N-1) \end{bmatrix}, \tag{4}$$

respectively. The m-th row and n-th column of $\mathbf{Y}'_k$ is expressed as [22]

$$\begin{aligned} y_k(m,n) = \sum_{i=0}^{N_t-1} \sum_{l=0}^{L_p-1} & \widetilde{f}_l \widetilde{F}_l \cdot e^{-j\ 2\pi(f_c+n\Delta f)\tau_l} \cdot e^{j2\pi f_{d,l} m T} \\ & \cdot e^{j2\pi \frac{d_a}{\lambda}(i-1)\sin(\theta_t^l)} \cdot e^{j2\pi \frac{d_a}{\lambda}(k-1)\sin(\theta_r^l)} \\ & \cdot x(m,n) + z_k(m,n), \end{aligned} \tag{5}$$

where L_p is the number of multiple paths, $\tau_l = \frac{r_l}{c}$ is the time delay, $f_{d,l} = \frac{v_l f_c}{c}$ is the Doppler frequency, c is the speed of light, λ is the wavelength of ISAC signal, $d_a = \frac{\lambda}{2}$ is the distance between the adjacent transmit and receive antenna elements, θ_t^l and θ_r^l are the angle of departure (AoD) and the angle of arrival (AoA) of the l-th path, respectively. The small scale fading factor $\widetilde{f}_l$ obeys the Weibull distribution [24]. The large scale fading factor $\widetilde{F}_l$ is expressed as $\widetilde{F}_l \approx \frac{\sigma_l \lambda^2}{(4\pi)^3 (r_{1,l} r_{2,l})^2}$, where σ_l is the radar cross section (RCS) of the l-th target, $r_{1,l}$ and $r_{2,l}$ are the range from the transmitter to the target and the target to the receiver, respectively. The range of the l-th path r_l is expressed as $r_{1,l} + r_{2,l}$. The matrix $\mathbf{X}' \in \mathcal{C}^{M \times N}$ is the transmitted OFDM signal, $\mathbf{Y}'_k \in \mathcal{C}^{M \times N}$ is the received OFDM signal, $\mathbf{Z}'_k \in \mathcal{C}^{M \times N}$ is the additive white Gaussian noise (AWGN) matrix.

Then, in the MIMO system, the m-th OFDM symbol of the received signal with the n-th subcarrier is composed into a vector $\mathbf{y}(m,n)$, which is expressed as

$$\mathbf{y}(m,n) = \sum_{l=0}^{L_p-1} \tilde{\mathbf{H}}^l(m,n)\mathbf{x}(m,n) + \mathbf{z}(m,n), \tag{6a}$$

$$\mathbf{x}(m,n) = [x(m,n), \cdots, x(m,n)]^T \in \mathcal{C}^{N_t \times 1} \tag{6b}$$

$$\mathbf{y}(m,n) = [y_0(m,n), \cdots, y_{N_r-1}(m,n)]^T \in \mathcal{C}^{N_r \times 1} \tag{6c}$$

$$\mathbf{z}(m,n) = [z_0(m,n), \cdots, z_{N_r-1}(m,n)]^T \in \mathcal{C}^{N_r \times 1}, \tag{6d}$$

where $\mathbf{y}(m,n)$ is the signal matrix received from N_r received antennas at the m-th symbol and n-th subcarrier. $\mathbf{x}(m,n)$ is the signal matrix transmitted from N_t transmit antennas at the m-th symbol and n-th subcarrier. $\mathbf{z}(m,n)$ is the AWGN vector at N_r received antennas. There are two main points need to consider in the ISAC channel:

- The geometric channel model with single-reflection is adopted as the ISAC channel in this paper. There are two main reasons for supporting the reasonableness of the single-reflection geometric channel model.
  1) Since the signal loses a lot of power after being reflected by scatterers, the power of a signal that has been reflected multiple times is much smaller than that of a direct signal and a single reflection. The above property is more pronounced in high frequency signals and is also present for low frequency signals. There are some related works that adopt the single-reflection geometric channel model in low frequency system, such as 3GHz in [18].
  2) For target sensing, the time delay and Doppler shift information carried by the received echo signals after multiple reflections do not truly present the distance and velocity information of the target, which in turn interferes with the estimation of target parameters. Therefore, the received echo signals after multiple reflections are usually regarded as sensing clutter signals. The research on sensing clutter suppression [25], [26] has been relatively mature and is not the focus of this paper.
- We consider ISAC scenarios where the environmental target is a point target rather than a extended target with large size. The main difference between the channel models for point targets and extended targets lies in the scattering characteristics. In the case of point targets, the received signal is mainly affected by a single dominant scatter, while in the case of extended targets, multiple scatterers contribute to the received signal. To accurately model the channel for extended targets, techniques such as ray tracing, geometric optics, or statistical channel models can be employed. These models take into account the scattering properties of the target and provide a more realistic representation of the channel response. Therefore, if the size of the target/scatter is large, the problem of how to define the spatial angles relative to the transceiver is more complex. For each extended target, the spatial angles relative to the transceiver can be expressed as a angle set $\{\theta_i | \theta_{\min} \le \theta_i \le \theta_{\max}\}$, where the angle region, $[\theta_{\min} \sim \theta_{\max}]$ is decided by the size of the target. ISAC signal processing for extended targets will be the focus of our future work.

Therefore, the channel matrix of the l-th path $\tilde{\mathbf{H}}^l(m,n) \in \mathcal{C}^{N_r \times N_t}$ can be expressed as

$$\begin{aligned} \tilde{\mathbf{H}}^l(m,n) = & \widetilde{f}_l \widetilde{F}_l \cdot e^{-j\ 2\pi(f_c+n\Delta f)\tau_l} \\ & \cdot e^{j2\pi f_{d,l} m T} \cdot \mathbf{d}_r(\theta_r^l)\mathbf{d}_t^H(\theta_t^l), \end{aligned} \tag{7}$$

where $\mathbf{d}_t$ and $\mathbf{d}_r$ are the transmitting and receiving steering vector, which can be expressed as

$$\mathbf{d}_t(\theta_t^l) = \frac{1}{\sqrt{N_t}} \left[1, e^{j2\pi \frac{d_a}{\lambda}\sin(\theta_t^l)}, \ldots, e^{j2\pi \frac{d_a}{\lambda}(N_t-1)\sin(\theta_t^l)}\right]^T$$

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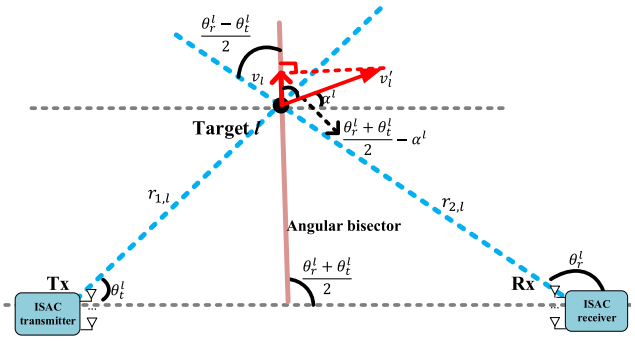


Fig. 3. Delay and Doppler frequency shift for passive sensing.

$$\mathbf{d}_r(\theta_r^l) = \frac{1}{\sqrt{N_r}}\left[1, e^{j2\pi\frac{d_a}{\lambda}\sin(\theta_r^l)}, \ldots, e^{j2\pi\frac{d_a}{\lambda}(N_r-1)\sin(\theta_r^l)}\right]^T. \tag{8}$$

Note that (6) can be written into a matrix form as

$$\mathbf{Y} = \sum_{l=0}^{L_p-1} \mathbf{H}_a^l \mathbf{X} \mathbf{H}_\varphi^l + \mathbf{Z}, \tag{9}$$

$$\mathbf{H}^l = \mathcal{S}(\mathbf{H}_a^l, \mathbf{H}_\varphi^l), \tag{10}$$

where

$$\begin{aligned} \mathbf{Y} &= [\mathbf{y}_1, \mathbf{y}_2, \cdots, \mathbf{y}_{N_r}]^T \in \mathcal{C}^{N_r \times MN}, \\ \mathbf{X} &= [\mathbf{x}, \mathbf{x}, \cdots, \mathbf{x}]^T \in \mathcal{C}^{N_t \times MN}, \\ \mathbf{Z} &= [\mathbf{z}_1, \mathbf{z}_2, \cdots, \mathbf{z}_{N_r}]^T \in \mathcal{C}^{N_r \times MN}, \end{aligned} \tag{11}$$

with

$$\begin{aligned} \mathbf{y}_k &= \mathrm{vec}(\mathbf{Y}_k') \in \mathcal{C}^{MN\times 1}, \\ \mathbf{x} &= \mathrm{vec}(\mathbf{X}') \in \mathcal{C}^{MN\times 1}, \\ \mathbf{z}_k &= \mathrm{vec}(\mathbf{Z}_k') \in \mathcal{C}^{MN\times 1}, \end{aligned} \tag{12}$$

where $\mathbf{x}$ is the transmitted signal, $\mathbf{y}_k$ is the received signal from the k-th antenna, $\mathbf{z}_k$ is the noise signal from the k-th antenna. The channel matrix of the l-th path $\mathbf{H}^l$ consists of two parts, i.e., channel amplitude information matrix $\mathbf{H}_a^l \in \mathcal{C}^{N_r \times N_t}$ and channel phase information matrix $\mathbf{H}_\varphi^l \in \mathcal{C}^{MN\times MN}$

$$\mathbf{H}_a^l = \widetilde{f}_l \widetilde{F}_l \cdot \mathbf{d}_r(\theta_r^l)\mathbf{d}_t^H(\theta_t^l), \tag{13}$$

$$\mathbf{H}_\varphi^l = \mathrm{diag}\left(h_\varphi^l(0,0), \cdots, h_\varphi^l(M-1, N-1)\right), \tag{14}$$

with

$$h_\varphi^l(m,n) = e^{-j\ 2\pi(f_c + n\Delta f)\tau_l} \cdot e^{j2\pi f_{d,l} m T}. \tag{15}$$

As shown in Fig. 3, the delay τ_l and Doppler frequency shift $f_{d,l}$ of the l-th target can be expressed as

$$\tau_l = \frac{r_l}{c} = \frac{r_{1,l} + r_{2,l}}{c}, \tag{16}$$

$$f_{d,l} = \frac{v_l}{\lambda} = -\frac{v_l'}{c} f_c \cdot \cos\left(\frac{\theta_r^l - \theta_t^l}{2}\right) \cdot \cos\left(\alpha^l - \frac{\theta_r^l + \theta_t^l}{2}\right), \tag{17}$$

where α^l is the angle of the l-th target velocity in the horizontal direction; v_l' is the absolute velocity of the l-th target; v_l is the radial velocity of the l-th target. Passive sensing can estimate the time delay and Doppler shift to get the r_l and v_l of the l-th target. The proof of (17) is provided in Appendix A.

III. ISAC Signal Processing

The signal processing of integrated passive sensing and communication includes two signal processing tasks: communication signal detection and passive sensing. Communication signal detection is to demodulate the transmitted signal data from the received signal. Passive sensing is to extract the range and velocity information contained in the channel information from the received signal. Based on the least square criterion, the error between the reconstructed received signal and the received signal is minimized. Therefore, the ISAC signal processing problem is formulated as

$$\mathcal{P}: \min_{\mathbf{X},[\mathbf{r},\mathbf{v}]} \left\| \mathbf{Y} - \sum_{l=0}^{L_p-1} \mathbf{H}_a^l \mathbf{X} \mathbf{H}_\varphi^l \right\|_2, \tag{18a}$$

$$\text{s.t.} \mathbf{H}^l = \mathcal{S}(\mathbf{H}_a^l, \mathbf{H}_\varphi^l) = \mathcal{F}_3([\mathbf{r}, \mathbf{v}]), \tag{18b}$$

where

$$\begin{aligned} \mathbf{r} &= \left[r_0, r_1, \cdots, r_l, \cdots, r_{L_p-1}\right]^T \\ \mathbf{v} &= \left[v_0, v_1, \cdots, v_l, \cdots, v_{L_p-1}\right]^T, \end{aligned} \tag{19}$$

are the estimated range and velocity vectors of targets, $\mathbf{X}$ is the detected transmitted signal. As shown in (18a), the objective of the optimization problem is to minimize the error between the reconstructed received signal and the real received signal $\mathbf{Y}$. To achieve the objective, there are two constraints: accurately demodulate the transmitted signal $\mathbf{X}$, accurately estimate the environmental targets and the communication channels. In (18b), the sensing results can be obtained by the 2D-DFT algorithm [5], [17], which is introduced in Section III – B. The ISAC channel reconstruction function $\mathcal{F}_3(\cdot)$ will be introduced in Section III – C.

According to (18), the signal processing of integrated passive sensing and communication is to simultaneously optimize the demodulated communication symbols and the detected target parameters to achieve the optimal communication and sensing performance. The problem solved by signal processing is a multi-parameter optimization problem, which includes the demodulated transmitted signal $\mathbf{X}$ and radar sensing results $[\mathbf{r}, \mathbf{v}]$. The iteration algorithm is a typical algorithm to solve the multi-parameter optimization problem [27]. Therefore, we propose an ISAC signal processing algorithm based on iteration, which consists of the following iterations

$$\mathbf{X}^{k+1} = \mathcal{F}_1\left(\mathbf{X}^k, \mathbf{H}^k\right), \tag{20a}$$

$$\left[\mathbf{r}^{k+1}, \mathbf{v}^{t+1}\right] = \mathcal{F}_2\left(\mathbf{X}^{k+1}, \mathbf{Y}\right), \tag{20b}$$

$$\mathbf{H}^{k+1} = \mathcal{F}_3\left(\mathbf{r}^{k+1}, \mathbf{v}^{k+1}\right), \tag{20c}$$

where k is the index of iterations, $\mathbf{H}$ represents the channel matrix set of L_p paths, $\mathcal{F}_1$, $\mathcal{F}_2$ and $\mathcal{F}_3$ are the optimization function of $\mathbf{X}$, $[\mathbf{r}, \mathbf{v}]$ and $\mathbf{H}$, which are listed in TABLE II.

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TABLE II
ITERATIONS IN ISAC SIGNAL PROCESSING

Function	Specific form	Section
$\mathcal{F}_1$: Signal detection	$\mathbf{X}_d^{k+1} = \eta_1^k \mathbf{X}_d^k + \sum_{l=0}^{L_p-1} \hat{\mathbf{H}}_a^l{}^T \mathbf{Z}_d^k \hat{\mathbf{H}}_\varphi^l{}^T$ $\mathbf{Z}_d^{k+1} = \eta_2^k \mathbf{Z}_d^k + \mathbf{Y}_d - \sum_{l=0}^{L_p-1} \hat{\mathbf{H}}_a^l \mathbf{X}_d^k \hat{\mathbf{H}}_\varphi^l$	III-A
$\mathcal{F}_2$: Passive sensing	$\mathbf{r}^{k+1} = \mathsf{IDFT}\left(\mathbf{X}^{k+1}, \mathbf{Y}\right)$ $\mathbf{v}^{k+1} = \mathsf{DFT}\left(\mathbf{X}^{k+1}, \mathbf{Y}\right)$	III-B
$\mathcal{F}_3$: Channel reconstruction	$\mathbf{H}^{k+1} = \mathsf{CHA}\left(\mathbf{r}^{k+1}, \mathbf{v}^{k+1}\right)$	III-C

A. $\mathcal{F}_1$: Improved D-AMP Signal Detection Algorithm

The signal detection function $\mathcal{F}_1$ is to demodulate transmitted data signal from the received signal. The input of $\mathcal{F}_1$ is $\mathbf{Y}_d$, $\mathbf{H}_a^l$, $\mathbf{H}_\alpha^l$, and the output of $\mathcal{F}_1$ is $\mathbf{X}_d^{k+1}$. Based on (9), the received data signal is expressed as

$$\mathbf{Y}_d = \sum_{l=0}^{L_p-1} \mathbf{H}_a^l \mathbf{X}_d \mathbf{H}_\varphi^l + \mathbf{Z}_d, \tag{21}$$

where $\mathbf{Z}_d$ is the AWGN on data signal. The maximum a posterior (MAP) is the typical algorithm for recovering the transmitted data signal $\mathbf{X}_d$ from the received data signal by maximizing the posterior probability $P\left(\mathbf{X}_d|\mathbf{Y}_d, \hat{\mathbf{H}}^l\right)$. However, the computational complexity of MAP is high due to the large number of matrix inversion operations required. To address this problem, D-AMP and OAMP algorithms have been proposed in [22] and [28], respectively. The D-AMP algorithm decouples the posterior probability $P\left(\mathbf{X}_d|\mathbf{Y}_d, \hat{\mathbf{H}}^l\right)$ into a series of $P\left(\mathbf{X}_d^k|\mathbf{Y}_d, \hat{\mathbf{H}}^l\right), (k = 1, 2, \ldots, K)$ in an iterative way, with K being the number of iterations. The estimated channel $\hat{\mathbf{H}}^l$ and the received data signal $\mathbf{Y}_d$ can be obtained in advance. According to the description of Algorithm 1 in [22], the OAMP algorithm consists of four main steps, which includes high complexity operations such as computing the expectation as well as the trace of a matrix. According to Algorithm 1, the D-AMP algorithm used in this paper only consists of two main steps, and does not require operations such as computing expectations. Therefore, the complexity of D-AMP is relatively small compared to OAMP.

The framework of typical D-AMP algorithms can be expressed as [29]

$$\begin{aligned} \mathbf{z}^{l+1} &= \mathbf{r} - \mathbf{W}\hat{\mathbf{h}}^{l+1} + \frac{1}{K}\mathbf{z}^l \, \mathrm{div} D_{\hat{\sigma}^l}\left(\hat{\mathbf{h}}^l + \mathbf{W}^T\mathbf{z}^l\right), \\ \hat{\mathbf{h}}^{l+1} &= D_{\hat{\sigma}^l}\left(\hat{\mathbf{h}}^l + \mathbf{W}^T\mathbf{z}^l\right), \end{aligned} \tag{22}$$

where $\hat{\mathbf{h}}^l$ is the input of the channel, $\mathbf{Z}^l$ is the input of the residual vector, $\hat{\sigma}^l$ denotes the parameter of the denoiser. The denoising function $D_{\hat{\sigma}^l}(\cdot)$ is a mathematical operation that removes noise from a signal, which is one of the key factors that determine the performance of D-AMP [28]. There are many related works on the denoising function, such as wavelet domain denoising [30]. However, these methods need to determine different hyper-parameters depending on particular scenarios. To solve the problem, we propose the improved D-AMP algorithm in Algorithm 1, which mainly contains two modules: signal updating (23) and noise updating (24). Different from the conventional D-AMP, the improved D-AMP sets two hyperparameters η_1 and η_2. Note that there are no convex/non-convex constraints in the improved D-AMP problem, we solve the problem in Algorithm 3 by setting the hyperparameters directly. However, the performance of the improved D-AMP algorithm heavily depends on the hyperparameter setting. To enhance the robustness of the algorithm, we further adopt the DL method to estimate the two hyperparameters, which will be introduced in Section IV.

Algorithm 1 Improved D-AMP Algorithm

Input: Received data signal $\mathbf{Y}_d$, estimated channel matrix $\hat{\mathbf{H}}^l = \mathcal{S}\left(\hat{\mathbf{H}}_a^l, \hat{\mathbf{H}}_\varphi^l\right)$.

1). Initialize: $\mathbf{Z}_d^1 \leftarrow \mathbf{0}$, $\mathbf{X}_d^1 \leftarrow \mathbf{0}$.

2). Update $\mathbf{X}_d^{k+1}$:

$$\mathbf{X}_d^{k+1} = \eta_1^k \mathbf{X}_d^k + \sum_{l=0}^{L_p-1} \hat{\mathbf{H}}_a^l{}^T \mathbf{Z}_d^k \hat{\mathbf{H}}_\varphi^l{}^T. \tag{23}$$

3). Update $\mathbf{Z}_d^{k+1}$:

$$\mathbf{Z}_d^{k+1} = \eta_2^k \mathbf{Z}_d^k + \mathbf{Y}_d - \sum_{l=0}^{L_p-1} \hat{\mathbf{H}}_a^l \mathbf{X}_d^k \hat{\mathbf{H}}_\varphi^l. \tag{24}$$

Output: Recovered data signal $\mathbf{X}_d^{K+1}$.

B. $\mathcal{F}_2$: 2D-DFT Passive Sensing Algorithm

The passive sensing function $\mathcal{F}_2$ is to extract the range and velocity information of the targets contained in the channel information. The input of $\mathcal{F}_2$ is $\mathbf{Y}$, $\mathbf{X}^{k+1}$, and the output of $\mathcal{F}_2$ is $\mathbf{r}^{k+1}$, $\mathbf{v}^{k+1}$. In terms of passive sensing algorithms, 2D-DFT is a typical sensing algorithm used in ISAC OFDM signal, which achieves relatively high sensing performance with low complexity [5]. The 2D-FFT based sensing algorithm aims to extract the distance and velocity information of targets from the received signal. Since the signal processing of passive sensing at different antennas is the same, without loss of generality, the signal processing algorithm of passive sensing at the k-th received antenna is introduced. Based on the signal model mentioned in Section II – B and Section II – C, the n-th subcarrier m-th OFDM symbol of the baseband received signal can be rewritten as [5]

$$\begin{aligned} y_k(m,n) = &\sum_{i=0}^{N_t-1} \sum_{l=0}^{L_p-1} A_{i\to k}(m,n) x(m,n) \\ &\cdot e^{-j2\pi n \Delta f \frac{\tau_l}{c}} \cdot e^{-\frac{j2\pi m T v_l f_c}{c}} \\ &+ z_k(m,n), \end{aligned} \tag{25}$$

$$\begin{aligned} A_{i\to k}(m,n) = &\widetilde{f}_l \widetilde{F}_l \cdot e^{j2\pi \frac{d_a}{\lambda}(i-1)\sin(\theta_t^l)} \\ &\cdot e^{j2\pi \frac{d_a}{\lambda}(k-1)\sin(\theta_r^l)}, \end{aligned} \tag{26}$$

where $z_k(m,n)$ is the AWGN, $x(m,n)$ is the transmitted modulation signal. The rectangular window function $\mathrm{rect}(\cdot)$ with time delay can be neglected in baseband processing [5]. The transmitted signal $x(m,n)$ can be obtained by improved

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D-AMP algorithm introduced in Section III – A. Then, after removing the transmitted information from the received information symbols by an element-wise complex division, the elements of division matrix expression can be derived as

$$\begin{aligned}(\mathbf{S}_g)_{m,n} &= \frac{y_k(m,n)}{x(m,n)} \\ &\approx A_{i\to k}(m,n)k_r(n)k_v(m),\end{aligned} \tag{27}$$

where

$$k_r(n) = e^{-j2\pi n\Delta f\frac{r_l}{c}}, \tag{28}$$

$$k_v(m) = e^{-\frac{j2\pi mTv_lf_c}{c}}. \tag{29}$$

Applying DFT for each row of $\mathbf{S}_g$, the velocity of target v_l can be deduced as follows

$$\begin{aligned}I^l_{s,n} &= \mathsf{DFT}(\mathbf{S}_g) \\ v_l &\in \left[\frac{I^l_{s,n}\cdot c}{TM_If_c}, \frac{\left(I^l_{s,n}+1\right)\cdot c}{TM_If_c}\right),\end{aligned} \tag{30}$$

where $I^l_{s,n}$ is the index of the peak of the DFT of the n-th row of $\mathbf{S}_g$, M_I is the number of DFT points. Applying inverse DFT (IDFT) for each column of $\mathbf{S}_g$, the range of target r_l can be deduced as follows

$$\begin{aligned}I^l_{s,m} &= \mathsf{IDFT}\left(\mathbf{S}_g\right) \\ r_l &\in \left[\frac{I^l_{s,m}\cdot c}{N_I\Delta f}, \frac{\left(I^l_{s,m}+1\right)\cdot c}{N_I\Delta f}\right),\end{aligned} \tag{31}$$

where $I^l_{s,m}$ is the index of the peak of the IDFT of the m-th column of $\mathbf{S}_g$, N_I is the number of IDFT points. Details of this method is described in [5]. However, the conventional 2D-DFT algorithm cannot achieve high sensing accuracy for multiple targets estimation. The reason is that the 2D-DFT algorithm estimates target distance and velocity information based on the peak index of DFT outputs, each target theoretically generates a peak and multiple parapets. When there are multiple targets, the side flaps generated by some targets will interfere with the peaks generated by other targets, thus affecting the accuracy of target parameter estimation. To address this problem, we make some improvements to the traditional 2D-DFT algorithm to obtain more accurate multi-target estimation results by cyclically extracting the peaks of the DFT output, the specific algorithm can be found in step 4 of Algorithm 2.

C. $\mathcal{F}_3$: ISAC Channel Reconstruction Algorithm

The ISAC channel reconstruction function $\mathcal{F}_3$ is to reconstruct the ISAC channel based on the passive sensing results. The input of $\mathcal{F}_3$ is the estimated range and velocity of targets $\mathbf{r}$, $\mathbf{v}$, and the output of $\mathcal{F}_3$ is $\mathbf{H}^{k+1}$. Since the range and velocity of the l-th target, i.e., r_l and v_l, can be obtained by passive sensing, we can reconstruct the ISAC channel $\mathsf{CHA}(\cdot)$ based on (10), (13), (14), (15), (16) and (17). Details of the ISAC channel reconstruction algorithm are presented in Algorithm 2.

Algorithm 2 ISAC Channel Reconstruction Algorithm

Input: Estimated range and velocity of targets $\mathbf{r}$, $\mathbf{v}$.

1). Calculate the AoD and AoA of targets by AoD and AoA estimation algorithm [16].

2). Calculate the delay $\boldsymbol{\tau}$ and Doppler frequency shift $\mathbf{f}_d$:

$$\begin{aligned}\boldsymbol{\tau} &= \frac{\mathbf{r}}{c}, \\ \mathbf{f}_d &= \frac{\mathbf{v}}{\lambda} = \frac{f_c}{c}\mathbf{v}.\end{aligned} \tag{32}$$

3). Reconstruct ISAC channel:

$$\hat{\mathbf{H}}^l = \mathsf{CHA}\left([\mathbf{r},\mathbf{v}]\right). \tag{33}$$

3a). Reconstruct the ISAC channel phase information $\hat{\mathbf{H}}^l_\varphi$ by $\boldsymbol{\tau}$ and $\mathbf{f}_d$ based on (14) and (15).

3b). Reconstruct the ISAC channel amplitude information matrix $\hat{\mathbf{H}}^l_a$ based on (13) and (8).

Output: Reconstructed ISAC channel matrix $\hat{\mathbf{H}}^l$ based on (10).

IV. ISAC-NET

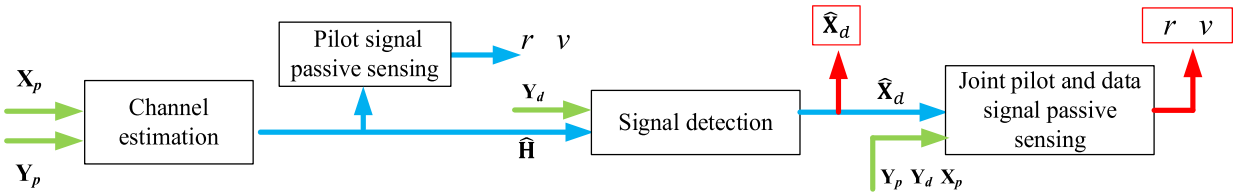
Although signal processing algorithm proposed in Section III improves the sensing and communication performance to a certain extent through iteration optimization, there are some hyper-parameters ($\Omega^k = \{\eta^k_1, \eta^k_2, \eta^k_3, \eta^k_4, \eta^k_5\}$) in Algorithm 3 that need to be set manually. The hyper-parameter configuration of the iterative optimization algorithm influences the performance. Improper hyper-parameter configuration degrades the performance and convergence. In this section, a model-driven DL network architecture, denoted by ISAC-NET, is constructed based on Algorithm 3, which learns the hyper-parameters in Algorithm 3 by DL to obtain better communication and sensing performance.

As shown in Fig. 4, the ISAC-NET has two modules:

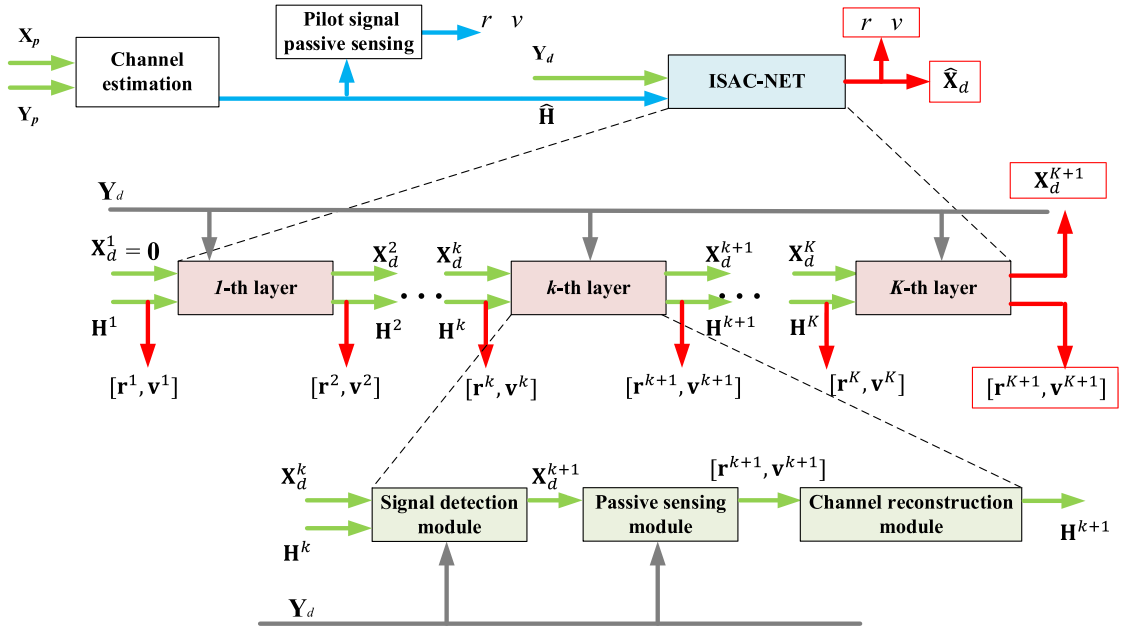
- Pilot signal processing: In this part, since both the transmitted pilot signal and the received pilot signal are known, the traditional channel estimation method can be used to obtain the channel information $\hat{\mathbf{H}}$ using the known pilot. Based on the pilot channel information, the 2D-FFT based passive sensing algorithm, can be used to complete the passive sensing and obtain the rough ranging and velocity measurement results, $[\mathbf{r},\mathbf{v}]$.
- ISAC-NET: In this part, we adopt the ISAC signal processing method based on model-driven DL. The ISAC-NET combines the conventional alternating optimization and deep unfolding into a neural network, which considers the key steps of ISAC signal processing, and designs three modules, including the passive sensing module, signal detection module and channel reconstruction module. The ISAC-NET can obtain suitable hyper-parameters by DL to guarantee the performance and convergence of communication and sensing signal processing.

In comparison to the data-driven DL-based joint channel estimation and signal detection architectures proposed in [31], [32], which use a large number of data to train the black-

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(a) Conventional architecture.



(b) ISAC-NET architecture.

Fig. 4. Conventional architecture $\mathcal{VS}$ ISAC-NET architecture. $\mathbf{H}$ represents the channel matrix set $\mathbf{H}^l$ of L_p paths; $\hat{\mathbf{H}}$ represents the estimated channel matrix; $\hat{\mathbf{X}}_d$ represents the signal matrix of the estimated transmitted signal.

box-based network, we construct the network architecture by employing the model-driven DL. Different from the model-driven DL-based joint channel estimation and signal detection architecture proposed in [22], which only optimizes the performance of communication signal detection without taking into account the performance of passive sensing. The ISAC-NET optimizes both communication and sensing performance. The structure of the ISAC-NET is illustrated in Fig. 4(b), which consists of K cascade layers and each layer has the same architecture but different trainable parameters. Each layer of ISAC-NET contains three main modules, the signal detection module, the passive sensing module and the channel reconstruction module, which are introduced in the following subsection.

Algorithm 3 ISAC signal processing algorithm

Input: Received data signal $\mathbf{Y}_d$, estimated channel matrix $\mathbf{H}^{l^k}$ and estimated data signal $\mathbf{X}_d^k$.

1). Initialize: $\mathbf{Z}_d^1 = \mathbf{0}$, $\mathbf{X}_d^1 = \mathbf{0}$.

2). Update $\mathbf{X}_d^{k+1}$:

$$\mathbf{X}_d^{k+1} = \eta_1^k \mathbf{X}_d^k + \sum_{l=0}^{L_p-1} \left(\hat{\mathbf{H}_a^l}^k\right)^T \mathbf{Z}_d^k \left(\hat{\mathbf{H}_\varphi^l}^k\right)^T. \quad (34)$$

3). Update $\mathbf{Z}_d^{k+1}$:

$$\mathbf{Z}_d^{k+1} = \eta_2^k \mathbf{Z}_d^k + \mathbf{Y}_d - \sum_{l=0}^{L_p-1} \hat{\mathbf{H}_a^l}^k \mathbf{X}_d^k \hat{\mathbf{H}_\varphi^l}^k. \quad (35)$$

while TRUE **do**

$[\mathbf{r}^{k+1}, \mathbf{v}^{k+1}] = [[\quad], [\quad]]$.

$\mathbf{H}_{new}^l = \mathbf{H}^{l^k} = \left\{\mathbf{H}_a^{l^k}, \mathbf{H}_\varphi^{l^k}\right\}$.

4). Passive sensing by 2D-DFT:

$$[M_v, i_v] = \mathrm{DFT}\left(\mathbf{X}_d^{k+1}, \mathbf{Y}_d\right), \quad (36)$$

$$[M_r, i_r] = \mathrm{IDFT}\left(\mathbf{X}_d^{k+1}, \mathbf{Y}_d\right), \quad (37)$$

where M_v and M_r are the maximum DFT and IDFT outputs, i_v and i_r are the index of the maximum DFT and IDFT outputs.

if $M_v \geq \eta_3^k$ or $M_r \geq \eta_3^k$ **then**

Obtain v_{tmp} and r_{tmp} based on (30) and (31).

$\mathbf{r}^{k+1} = \left[\mathbf{r}^{k+1}, r_{tmp}\right]$.

$\mathbf{v}^{k+1} = \left[\mathbf{v}^{k+1}, v_{tmp}\right]$.

Reconstruct ISAC channel:

$$\mathbf{H}_{tmp}^l = \mathrm{CHA}\left(r_{tmp}, v_{tmp}\right), \quad (38)$$

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$$\mathbf{H}_{new}^{l} = \mathbf{H}^{l^k} - \mathbf{H}_{tmp}^{l}. \tag{39}$$

else
break.
end if
end while

5). Update $\left[\mathbf{r}^{k+1}, \mathbf{v}^{k+1}\right]$:

$$\left[\mathbf{r}^{k+1}, \mathbf{v}^{k+1}\right] = \eta_4^k \left[\mathbf{r}^{k}, \mathbf{v}^{k}\right] + \left[\mathbf{r}^{k+1}, \mathbf{v}^{k+1}\right]. \tag{40}$$

6). Reconstruct the ISAC channel:

$$\mathbf{H}^{l^{k+1}} = \mathsf{CHA}\left(\left[\mathbf{r}^{k+1}, \mathbf{v}^{k+1}\right]\right). \tag{41}$$

7). Update $\mathbf{H}^{l^{k+1}}$:

$$\mathbf{H}^{l^{k+1}} = \eta_5^k \mathbf{H}^{l^k} + \mathbf{H}^{l^{k+1}}. \tag{42}$$

Output: The recovered data signal $\mathbf{X}_d^{K+1}$, estimated range and velocity of targets $\left[\mathbf{r}^{K+1}, \mathbf{v}^{K+1}\right]$.

A. Each Layer of ISAC-NET

As Fig. 5 shows, for the k-th layer of the ISAC-NET:

- The input is the estimated data signal $\mathbf{X}_d^k$, the received data signal $\mathbf{Y}_d$, the estimated channel matrix $\mathbf{H}^k$ and the noise matrix $\mathbf{Z}^k$ from the $(k-1)$-th layer.
- The ISAC signal processing module of each layer is performed as in Algorithm 3, which is developed by unfolding the improved D-AMP algorithm, the 2D-DFT algorithm and the ISAC channel reconstruction algorithm.
- The blue part, the green part and the yellow part denote the improved D-AMP algorithm, the 2D-DFT based passive sensing algorithm and the ISAC channel reconstruction algorithm.

Compared with the conventional 2D-DFT passive sensing algorithm, the ISAC signal processing algorithm realizes the multi-target sensing by alternating between the estimation of target parameters and channel reconstruction. This method has certain improvement in the performance of multiple targets sensing, which is verified in the subsequent simulation analysis. Moreover, there are five main learnable variables $\Omega^k = \left\{\eta_1^k, \eta_2^k, \eta_3^k, \eta_4^k, \eta_5^k\right\}$ in each layer, which is much less than the data-driven DL [31], [33], greatly reducing the complexity of network training.

B. Model Training

Details of the ISAC-NET training parameters are introduced in this subsection, which consists of three main parts: data generation, training settings, training and deployment.

- **Data generation:** The inputs of ISAC-NET is the received signal, whilst the outputs of ISAC-NET is the demodulated transmitted signal and the estimated target velocity and target distance. The training and testing data sets are obtained through MATLAB simulation, in which 95% of the data is used for training and 5% of the data is used for testing. The data set includes the transmitted signal set and the received signal set under different signal-to-noise ratio (SNR) conditions, from $-20 \sim 60$ dB. In each model training, a random data sequence is first generated as the transmitted symbols and the corresponding OFDM frame is formed with a sequence of pilot symbols and the pilot symbols need to be fixed during the training stages. Specifically, the transmitted signal is obtained by (1), the received signal is obtained by (6), the channel matrix is obtained by (10), (13) and (14).
- **Training settings:** Each batch of training data is set up as an OFDM signal matrix containing 1024 subcarriers and 256 OFDM symbols. The batch size of each training epoch is 128, and the number of training epochs is 500. The learning rate is set to 0.001.
- **Training and deployment:** Based on the offline training, we obtain the optimized parameters $\mathbf{\Omega} = \left\{\Omega^k\right\}_{k=1}^K$ for different SNRs using the tensorflow platform. The optimized parameters are stored to detect the modulated symbols in the deployment stage. The ISAC-NET can be interpreted as a new iterative detector after training. The incorporated learnable parameters can be adaptively adjusted according to the practical channels, real symbol detection errors and passive sensing errors, improving the performances of both communication and sensing.

C. Complexity Analysis

In this subsection, we analyze the computational complexity of Algorithm 3 and ISAC-NET. Without loss of generality, we focus on the complex multiplication operations in the algorithm and ignore the less computationally complex addition operation. Note that both Algorithm 3 and ISAC-NET consists of some similar operation, Algorithm 3 includes some iterations, ISAC-NET includes some layers. Therefore, we only need to analyze the computational complexity of each iteration in Algorithm 3 and each layer in ISAC-NET. Each iteration in Algorithm 3 includes three main part:

- Improved D-AMP: As mentioned in Algorithm 1, there are two operations in the improved D-AMP: (23) and (24). Since the size of $\mathbf{H}_a^l$ is $N_r \times N_t$, the size of $\mathbf{Z}_d^k$ is $N_r \times MN$, the size of $\mathbf{H}_\varphi^l$ is $MN \times MN$, the complexity of (23) can be deduced as $\mathcal{O}(L_p N_t N_r (MN)^2)$. Similarly, the complexity of (24) can be deduced as $\mathcal{O}(L_p N_t N_r (MN)^2)$. Therefore, the complexity of the improved D-AMP can be deduced as $\mathcal{O}(2L_p N_t N_r (MN)^2)$.
- Passive sensing by 2D-DFT: According to [5], the complexities of typical 2D-DFT is $\mathcal{O}(Nlog(N)+Mlog(M))$. To achieve high sensing accuracy for multiple targets estimation, passive sensing by 2D-DFT, i.e. step 4) in Algorithm 3, includes L_p cycles of 2D-DFT operations. Therefore, the complexity of passive sensing can be deduced as $\mathcal{O}(L_p(Nlog(N) + Mlog(M)))$.
- ISAC channel reconstruction: As mentioned in Algorithm 2, each ISAC channel reconstruction consists of (8), (13), (14) and (15). According to step 4) in Algorithm 3, there are L_p cycles of the ISAC channel reconstruction. Therefore, the complexity of

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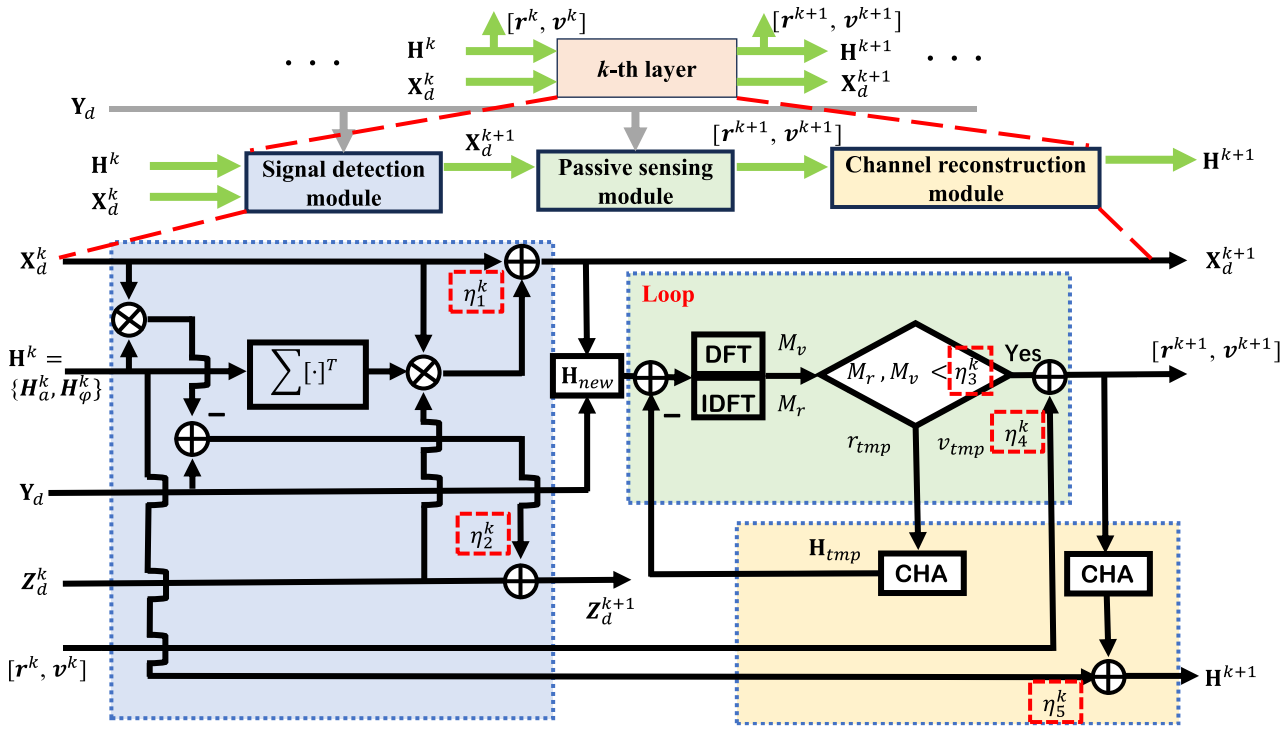


Fig. 5. Each layer of ISAC-NET. The blue boxed part on the left indicates the signal detection module, corresponding to the steps 1)-3) in Algorithm 3. The green boxed part on the upper right indicates the passive sensing module, corresponding to the steps 4)-5) in Algorithm 3. The yellow boxed part on the down right indicates the channel reconstruction module, corresponding to the steps 6)-7) in Algorithm 3.

the ISAC channel reconstruction can be deduced as $\mathcal{O}(L_p(N_tN_r + MN))$.

Assume that the iteration of Algorithm 3 is I_3, the layer of ISAC-NET is K, the overall complexity of Algorithm 3 can be expressed as $\mathcal{O}(I_3L_p(N_tN_r + MN + Nlog(N) + Mlog(M) + 2N_tN_r(MN)^2))$, the overall complexity of ISAC-NET can be expressed as $\mathcal{O}(KL_p(N_tN_r + MN + Nlog(N) + Mlog(M) + 2N_tN_r(MN)^2))$.

V. Simulation Results

To demonstrate the relationship between the communication demodulation error and sensing accuracy in the ISAC system, we first analyze the communication performance with sensing errors and the sensing performance in the condition of communication demodulation errors. To verify the improvement of ISAC-NET in the communication and sensing performance, the communication and sensing performance of ISAC-NET is simulated and compared with the typical communication and sensing signal processing algorithms. It should be noted that, the SNR adopted in the simulation denotes the SNR of the received signal, and each point in the curve is calculated over 500 Monte Carlo trials. Simulation parameters used in this section are shown in TABLE III [22], [23], [34].

TABLE III
Simulation Parameters

Items	Value	Meaning of the parameter
f_c	4 GHz	Carrier frequency
Δf	120 kHz	Carrier frequency spacing
M	256	Number of OFDM symbols
M_I	2560	Number of DFT points
N	1024	Number of subcarriers
N_I	10240	Number of IDFT points
T_p	8.3 μs	OFDM symbol period
T_c	2.08 μs	CP period
T	10.38 μs	The whole OFDM period
B	123 MHz	Frequency bandwidth
N_t	8	Number of transmitting antenna array
N_r	8	Number of receiving antenna array
L	$1 \sim 3$	Number of targets
σ	1 m^2	RCS of targets
16 QAM		Modulation type
SNR	$-20 \sim 60$ dB	SNR of received signals
w_p	1/14	Pilot signal length ratio

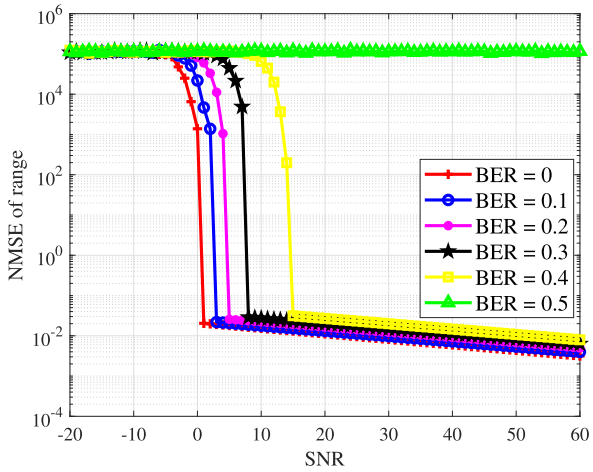
A. Communication and Sensing Performance Under Error Conditions

In this section, we analyze and simulate the communication and sensing performance in the condition of communication demodulation errors. Fig. 6 presents the normalized mean squared errors (NMSEs) of range and velocity with different SNRs, which can be expressed as

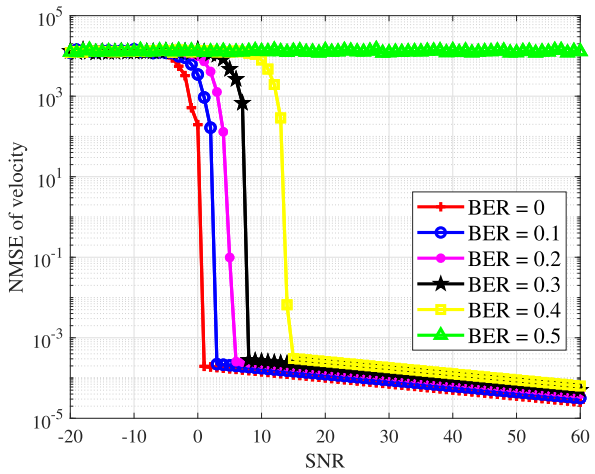
$$\Gamma_r = \frac{1}{L}\sum_{l=0}^{L-1}\frac{|\hat{r}_l - r_l|^2}{r_l^2},$$
$$\Gamma_v = \frac{1}{L}\sum_{l=0}^{L-1}\frac{|\hat{v}_l - v_l|^2}{v_l^2}, \quad (43)$$

where L is the number of targets, r_l and v_l are the actual range and velocity of the l-th target, $\hat{r}_l$ and $\hat{v}_l$ are the estimated range and velocity of the l-th target.

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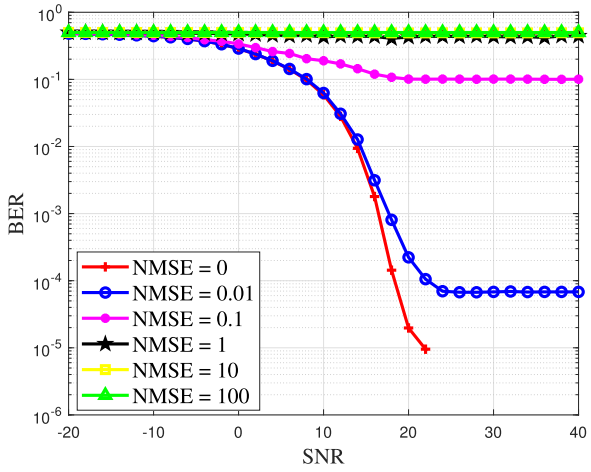


(a) NMSE of range.

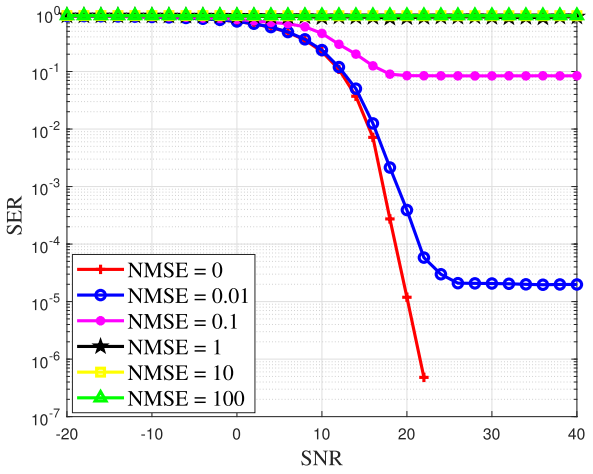


(b) NMSE of velocity.

Fig. 6. RMSE of range and velocity with different BERs.



(a) BER.



(b) SER.

Fig. 7. BER and SER with different RMSEs.

As shown in Fig. 6(a) and Fig. 6(b), under the same bit error rate (BER) condition, the NMSEs of range and velocity decrease with the increase of SNR, which means that the accuracy of passive sensing is improved. In addition, under the same SNR condition, the NMSEs of range and velocity increase with the increase of BER, which means that the passive sensing accuracy is decreased. When the BER reaches 0.5, it indicates that the recovered communication signal is completely wrong. At this time, the NMSEs of range and velocity do not decrease with the increase of SNR, and the sensing accuracy can no longer decrease. Therefore, we conclude that the accurate demodulation of the communication symbols is one of the preconditions for obtaining high-accurate passive sensing performance.

Fig. 7 presents the BER and symbol error rate (SER) versus different SNRs. As Fig. 7(a) and Fig. 7(b) show that the BER and SER decrease with the increase of SNR under the same NMSE condition, which means that the accuracy of communication demodulation is improved. In addition, under the same SNR condition, the BER and SER increase with the increase of NMSE, which means that the accuracy of communication demodulation is decreased. When the NMSE reaches 100, the BER is close to 0.5, which indicates that the recovered communication signal is completely wrong. Therefore, we conclude that accurate passive sensing is one of the preconditions for obtaining high-accurate communication performance.

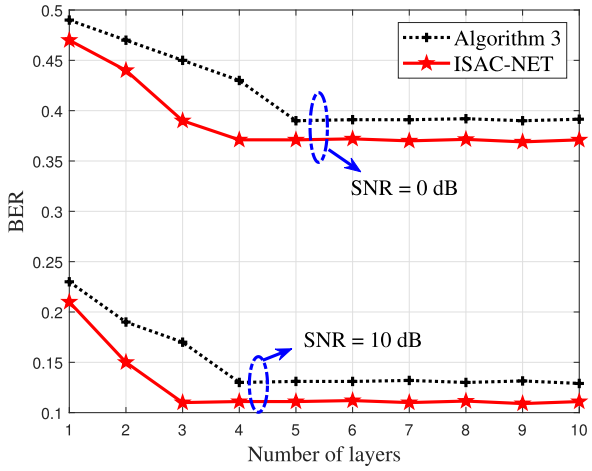
The simulation results provide a theoretical basis for the proposed ISAC-NET based on communication and sensing iterative optimization.

B. Communication and Sensing Performance of ISAC-NET

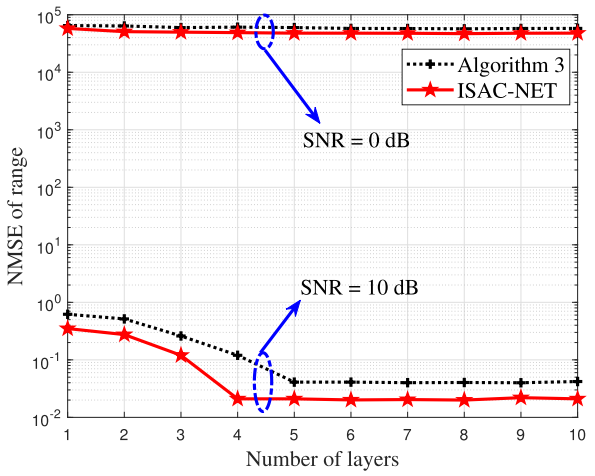
In this subsection, we analyze and simulate the communication and passive sensing performance of ISAC-NET. Since convergence is a necessary condition of the iterative algorithm, we first simulate and verify the convergence of ISAC-NET for sensing and communication. After that, the communication and sensing performance of the proposed ISAC-NET is simulated and compared with the typical communication and sensing signal processing algorithms.

1) Convergence Analysis: First, we analyze the convergence performance of Algorithm 3 and ISAC-NET. Fig. 8 illustrates

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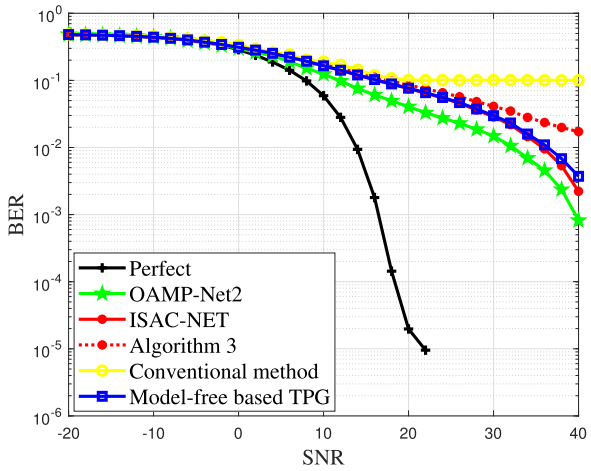


(a) BER.

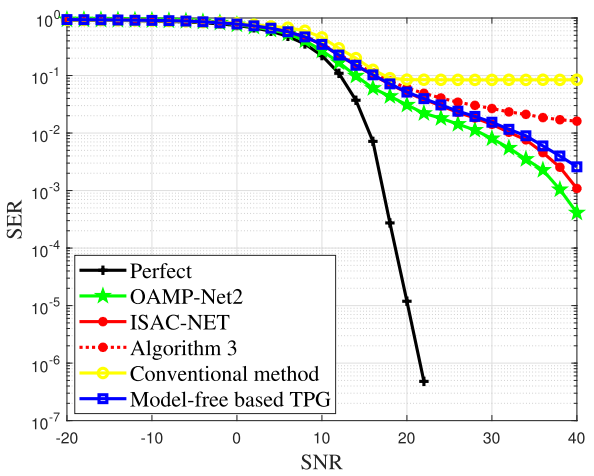


(b) NMSE of range.

Fig. 8. BER and NMSE of the ISAC-NET and Algorithm 3 versus the number of layers with different SNRs.



(a) BER.



(b) SER.

Fig. 9. BER and SER comparison of the ISAC-NET with other typical algorithms.

the BER and NMSE versus the number of layers (iterations) under SNR $=$ 0 dB and SNR $=$ 10 dB. From Fig. 8(a), Algorithm 3 convergences within five layers (iterations) for all the cases, ISAC-NET converges faster than Algorithm 3, within four layers (iterations) for all the cases. From Fig. 8(b), both the Algorithm 3 and ISAC-NET converge within five layers (iterations) for all the cases when SNR $=$ 10 dB. Moreover, the estimation error of ISAC-NET is smaller than that of Algorithm 3. Since SNR $=$ 0 dB does not meet the requirement of passive sensing, it is difficult to achieve high-accurate target detection, both the Algorithm 3 and ISAC-NET cannot converge. We can conclude that the SNR of input received signal will effect the convergence of the Algorithm 3 and the trained neural network, ISAC-NET.

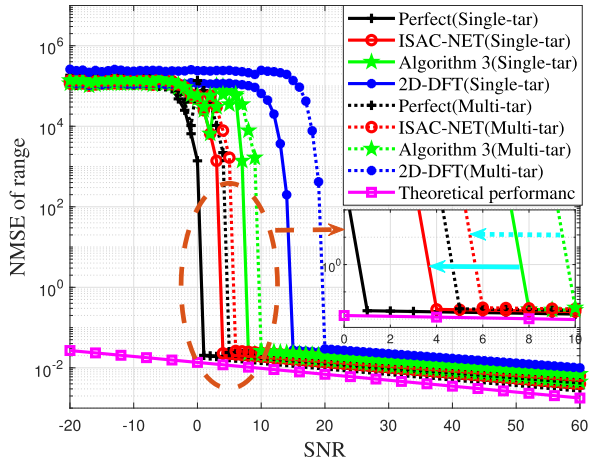
2) Performance Comparison: Through simulations, the communication and sensing performance of the proposed ISAC-NET is evaluated and compared to the performance of typical communication and sensing signal processing algorithms, which are listed in TABLE IV. For the multi-targets detection scenario, 3 targets are set to be detected. The actual range and velocity of targets are 10 m, 50 m, 100 m and 5 m/s, 10 m/s, 15 m/s, respectively.

Fig. 9 compares the BER and SER of MAP [22], OAMP-Net2 [22], Algorithm 3, and ISAC-NET. The perfect line denotes the BER and SER without CSI. The conventional method refers to the MAP method, which directly demodulates the received signal with channel estimation error. As Fig. 9 shows, ISAC-NET and OAMP-Net2 have similar BER and SER, smaller than that of Algorithm 3, which verifies the communication performance gain of deep unfolding over conventional algorithm without unfolding. Moreover, Fig. 9 compares the BER and SER performance of the ISAC-NET based on model-driven DL and the trainable projected gradient (TPG) algorithm [33] based on model-free DL. As shown in Fig. 9, the square blue line denotes the BER and SER performance of TPG, which is bigger than the star green line and close to the point red solid line. It means that the signal processing based on model-driven DL, OAMP-Net2 and ISAC-NET can achieve slightly higher BER performance than the signal processing based on model-free DL, TPG.

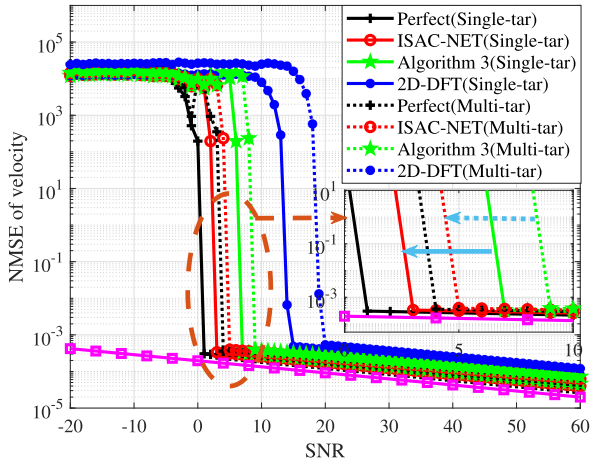
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TABLE IV
COMMUNICATION AND SENSING ALGORITHMS FOR SIMULATION ANALYSIS

Method	Communication	Sensing
Perfect	MAP without pilot CSI error	2D-DFT [5] without communication demodulation error
ISAC-NET	ISAC-NET with pilot CSI error	ISAC-NET with communication demodulation error
OAMP-Net2 [22]	OAMP-Net2 proposed in with pilot CSI error	
Conventional method	MAP with pilot CSI error	
2D-DFT		2D-DFT [5] with communication demodulation error



(a) NMSE of range.



(b) NMSE of velocity.

Fig. 10. NMSE comparison of the ISAC-NET with other typical algorithms.

Fig. 10 compares the NMSE of the 2D-DFT algorithm [5] and ISAC-NET. The perfect line denotes the NMSE without communication demodulation errors. As Fig. 10 shows, under the same SNR condition, NMSE of range and velocity by ISAC-NET is smaller than that of 2D-DFT based method. It means that ISAC-NET can obtain passive sensing performance with higher accuracy than 2D-DFT. Moreover, compared with the ISAC signal processing algorithm without unfolding (Algorithm 3), ISAC-NET obtains smaller NMSE of range and velocity, which demonstrates performance gain of deep unfolding over conventional sensing algorithm without unfolding.

The square pink line in Fig. 10(a) denotes the theoretical performance of range estimation, which can be deduced by the standard deviation of CraméRao bound, $\frac{\Delta R}{\sqrt{(2\mathrm{SNR})}}$ [35], with $\Delta R = \frac{c}{2B}$ being the ranging resolution. The square pink line in Fig. 10(b) denotes the theoretical performance of velocity estimation, which can be deduced by the standard deviation of CraméRao bound, $\frac{\Delta V}{\sqrt{(2\mathrm{SNR})}}$, with $\Delta V = \frac{c}{2TMf_c}$ being the velocity resolution. We can see that the red and black solid line is close to the square pink line when $\mathrm{SNR} \geq 0$ dB, which means that the sensing performance of range and velocity estimation is close to the theoretical performance in high SNR regions.

VI. CONCLUSION

In this paper, we proposed an integrated sensing and communication network based on model-driven DL, called ISAC-NET, which combines passive sensing with communication demodulation signal processing using model-driven DL. Dissimilar to existing passive sensing algorithms that carried out the communication signal detection and passive sensing in serial. The ISAC-NET obtains passive sensing results and communication demodulation simultaneously by employing the block-by-block signal processing method. The ISAC-NET was divided into the passive sensing module, signal detection module and channel reconstruction module. Moreover, the performance improvements of communication and sensing in ISAC-NET, such as the estimation accuracy of range and velocity estimation, BER and SER, are analyzed and simulated. The simulation results verified that the ISAC-NET obtains better performance in communication and passive sensing than that of the traditional signal demodulation algorithms. Compared to the 2D-DFT based sensing algorithm, ISAC-NET demonstrates significantly enhanced sensing performance.

APPENDIX A
DOPPLER FREQUENCY SHIFT FOR PASSIVE SENSING

As Fig. 3, the l-th target is moving with velocity $v_l{}'$ in the direction α^l. The carrier frequency of the ISAC signal at Tx is f_c. Based on the source reference frame of relativistic Doppler effects [36], the frequency of the ISAC signal arriving at the l-th target can be expressed as

$$f_1 = f_c \frac{1 - \beta \cos\left(\alpha^l - \theta_t^l\right)}{\sqrt{1-\beta^2}}, \tag{44}$$

where $\beta = \frac{v_l{}'}{c} \ll 1$ with c being the velocity of light. Similarly, the frequency of the ISAC signal at Rx can be given

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by

$$f_2 = f_1 \frac{\sqrt{1-\beta^2}}{1+\beta\cos\left(\alpha^l-\theta_r^l\right)}. \tag{45}$$

The Doppler frequency shift of the l-th target can be expressed as

$$\begin{aligned} f_{d,l} &= f_2 - f_c \\ &= f_c\left(\frac{\sqrt{1-\beta^2}}{1+\beta\cos\left(\alpha^l-\theta_r^l\right)}\cdot\frac{1-\beta\cos\left(\alpha^l-\theta_t^l\right)}{\sqrt{1-\beta^2}}-1\right) \\ &= f_c\left(\frac{1-\beta\cos\left(\alpha^l-\theta_t^l\right)}{1+\beta\cos\left(\alpha^l-\theta_r^l\right)}-1\right) \\ &= -\beta f_c\frac{\cos\left(\alpha^l-\theta_t^l\right)+\cos\left(\alpha^l-\theta_r^l\right)}{1+\beta\cos\left(\alpha^l-\theta_r^l\right)}. \end{aligned} \tag{46}$$

Based on $\beta = \frac{{v_l}'}{c} \ll 1$, the Doppler frequency shift of the l-th target $f_{d,l}$ can be approximated as

$$f_{d,l} \approx -\frac{{v_l}'}{c} f_c \cdot \cos\left(\frac{\theta_r^l-\theta_t^l}{2}\right)\cdot\cos\left(\alpha^l-\frac{\theta_r^l+\theta_t^l}{2}\right). \tag{47}$$

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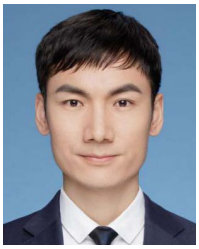
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